# RGB BULB
## Using NE555 Timer

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Red-green-blue (RGB) multi-colour bulbs available in the market are expensive as these are based on a microcontroller. The program for the microcontroller is difficult to understand. Here is a simple and inexpensive circuit for an RGB bulb using NE555 timer.

An RGB bulb is made up of three individual LEDs: red, green and blue. The intensities of these LEDs are varied to generate lights of different colours or shades.

The main objective of this project is to produce different light colours with the help of NE555 timer IC. The author's prototype is shown in Fig. 1.

### Circuit and working

Circuit diagram of the RGB bulb using NE555 timer is shown in Fig. 2. It is built around 230V AC primary to 12V-0-12V, 500mA secondary transformer (X1), a 5V regulator IC 7805 (IC1), three NE555 timers (IC2 through IC4), red, green and blue LEDs (LED1 through LED3), and a few other components.

To derive regulated 5V for the main circuit, 230V AC mains supply is stepped down by transformer X1, rectified by a full-wave rectifier comprising diodes D1 and D2 (1N4007), filtered by capacitor C1 and fed to IC1.

Three similar astable multivibrators are built around NE555 ICs (IC2, IC3 and IC4), each for red, green and blue LEDs, respectively. These use a 1-kilo-ohm resistor, a 10-kilo-ohm potmeter and a 0.1µF capacitor as R-C timing components.

Pins 1 and 8 of each NE555 are connected to ground and +5V, respectively. Reset input (pin 4) is also connected to +5V. Control voltage input (pin 5) is connected to ground through 0.01µF capacitor. 1-kilo-

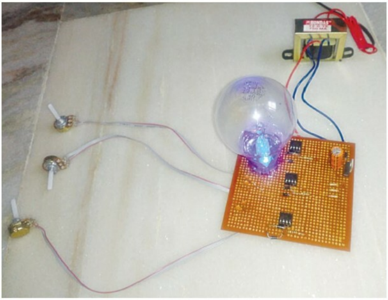

Fig. 1: Author's prototype

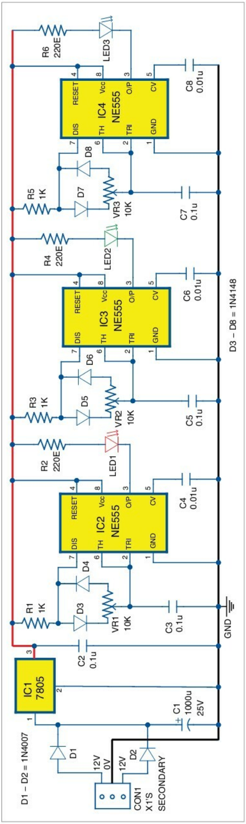


Fig. 2: Circuit diagram for the RGB bulb using NE555 timer